Penguin Data Analysis Report

Part A: Data Preparation

- **Sample Size:** The final sample size used in the analysis after removing missing observations is 333 penguins.

Part B: Exploratory Data Analysis

- **Highest Average Body Mass:** The Gentoo species has the highest average body mass at 5092.44 g.
- **Greater average body weight according to sex:** Males have higher average body weight (4545.69 g) than females (3862.27 g).
- **Difference by species:** The difference in body weight between male and female penguins is constant although the amount varies a little from one species to another. (e.g., the male–female gap is 674.66 g for Adelies, but only 411.76 g for Chinstraps).

Part C: One-Way ANOVA

- **Null Hypothesis (H_0):** Mean body mass is equal across all penguin species ($\mu_{\text{Adelie}} = \mu_{\text{Chinstrap}} = \mu_{\text{Gentoo}}$).
- **Alternative Hypothesis (H_a):** At least one species has a significantly different mean body mass.
- **ANOVA Results:** F-statistic = 341.9, Degrees of Freedom = 2 (model) and 330 (residuals), p-value $< 2 \times 10^{-16}$.
- **Statistical Decision:** Because the p-value is < 0.05, we reject the null hypothesis.
- **Interpretation:** There is overwhelming statistical evidence that penguin body mass differs significantly among the Adelie, Chinstrap, and Gentoo species.

Part D: Dummy-Variable Regression

- **Reference Category:** R alphabetically assigned Adelie as the baseline reference category.
- **Dummy Variables:** `DChinstrap` and `DGentoo`.

- **Regression Coefficients:**
  - **Intercept (3706.16):** The estimated mean body mass of an Adelie penguin.
  - **speciesChinstrap (26.92):** A Chinstrap penguin is estimated to weigh 26.92 g more than an Adelie penguin, on average (not statistically significant, $p = 0.691$).
  - **speciesGentoo (1386.27):** A Gentoo penguin is estimated to weigh 1386.27 g more than an Adelie penguin, on average ($p < 2 \times 10^{-16}$).
- **Relationship to ANOVA:** The regression yields the exact same overall F-statistic (341.9), p-value, and degrees of freedom as the One-Way ANOVA. While the ANOVA tests for any overall difference in means, the dummy regression explicitly compares the mean of each specific category against the Adelie baseline.

Part E: Two-Way ANOVA

- **Species Main Effect:** Sum of Squares = 145,190,219; df = 2; Mean Square = 72,595,110; $F = 758.358$; $p < 2 \times 10^{-16}$. We reject the null hypothesis. Mean body mass is not equal across species.
- **Sex Main Effect:** Sum of Squares = 37,090,262; df = 1; Mean Square = 37,090,262; $F = 387.460$; $p < 2 \times 10^{-16}$. We reject the null hypothesis. Mean body mass is not equal between sexes.
- **Species × Sex Interaction:** Sum of Squares = 1,676,557; df = 2; Mean Square = 838,278; $F = 8.757$; $p = 0.000197$. We reject the null hypothesis that there is no interaction. The effect of sex on body mass depends on the species.

Part F: Interaction Analysis

- **Parallel Profiles:** Because the interaction is highly significant ($p = 0.000197$), the profiles on the interaction plot are non-parallel.
- **Meaning of Non-parallel Profiles:** This means that the difference in mass between males and females is not the same in the three species.
- **Statistical Significance:** It is statistically significant that the interaction between the Species and Sex is significant.
- **Difference Across Species:** The difference in the body mass of males and females varies with species.
- **Practical Significance:** There is always a sexual dimorphism between males and females. For instance, the weight penalty for being female is much less severe in Chinstrap penguins compared to Adelie or Gentoo penguins.

Part G: Full Regression Model

- **Intercept (3368.84):** The estimated mean body mass for the reference group: a Female Adelie penguin.
- **Species Coefficients (Chinstrap = 158.37, Gentoo = 1310.91):** The estimated weight differences for females of these species compared to female Adelies.
- **Sex Coefficient (male = 674.66):** The estimated weight difference between male and female Adelies.
- **Interaction Coefficients (Chinstrap:male = −262.89, Gentoo:male = 130.44):** These represent the adjustment to the male weight gap for other species. For example, the difference between male and female Chinstraps is 262.89 g less than the difference between male and female Adelies.

Final Interpretation

The mass of penguins varies greatly depending on the species and the sex of the individual, and the role of a species in the weight of a penguin strictly depends on the sex of the penguin. The information observed and the statistical data confirm that Gentoo penguins are the heaviest species and males are heavier than females in all studied groups. Nevertheless, the practical interpretation of that effect ($p = 0.000197$) shows that there is no uniform sexual dimorphism. The weight gap between males and females shrinks significantly for Chinstrap penguins compared to Adelies and Gentoos, meaning one cannot accurately predict a penguin's body mass using species and sex as purely independent, additive factors.